CS23334 Fundamentals of Data Science Lab

Experiment 19

Exp No: 19 Experiment to understand Logistic Regression for a given data set.

Description: Understand the Logistic Regression algorithm for the dataset given.

Aim: To build, train, optimize, and evaluate a Logistic Regression classification model on Social_Network_Ads.csv to predict customer purchasing decisions based on Age and EstimatedSalary.

Methodology:

1 **Data Acquisition & Loading:** Load Social_Network_Ads.csv dataset containing 400 entries with attributes User ID, Gender, Age, EstimatedSalary, and target Purchased. Inspect DataFrame structure using df and df.head().

2 **Feature Extraction:** Extract independent numerical attributes (Age, EstimatedSalary) into features matrix using df.iloc[:, [2,3]].values and target binary label (Purchased) into label vector using df.iloc[:, 4].values.

3 **Random State Partition Tuning:** Iteratively partition dataset across random seeds (1 to 400) using train_test_split(test_size=0.2) to evaluate partitions where test accuracy outperforms training accuracy.

4 **Model Fitting:** Select optimal partition (random_state=286) and fit LogisticRegression() model from sklearn.linear_model on (x_train, y_train).

5 **Accuracy Evaluation:** Compute classification accuracy scores on training set (83.44%) and testing set (91.25%) using finalModel.score().

6 **Classification Metrics & Decision Boundary:** Generate classification report via classification_report() to measure precision, recall, and f1-score across non-purchasers (257 samples) and purchasers (143 samples), and visualize decision boundary.

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df

df.head()

features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features

label

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

for i in range(1,401):
    x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_stat
e=i)
    model=LogisticRegression()
```

```
    model.fit(x_train,y_train)
    train_score=model.score(x_train,y_train)
    test_score=model.score(x_test,y_test)
    if test_score>train_score:
        print("Test {} Train{} Random State {}".format(test_score,train_score,i))

x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=28
6)
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)

print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))

from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

OUTPUT: 1. Dataset Preview & Feature/Label Arrays

```
In [1]: import numpy as np
        import pandas as pd
        df=pd.read_csv('Social_Network_Ads.csv')
        df
Out[1]: 400 rows x 5 columns
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [4]: features=df.iloc[:,[2,3]].values
        label=df.iloc[:,4].values
        features
Out[4]: array([[   19, 19000],
               [   35, 20000],
               [   26, 43000],
               [   27, 57000],
               [   19, 76000], ...])

In [5]: label
Out[5]: array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1,
               1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
               ...], dtype=int64)
```

OUTPUT: 2. Random State Loop & Model Fitting

```
In [7]: for i in range(1,401):
            x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_s
tate=i)
            model=LogisticRegression()
            model.fit(x_train,y_train)
```

```
            train_score=model.score(x_train,y_train)
            test_score=model.score(x_test,y_test)
            if test_score>train_score:
                print("Test {} Train{} Random State {}".format(test_score,train_score,i))

        Test 0.6875 Train0.63125 Random State 3
        Test 0.7375 Train0.61875 Random State 4
        Test 0.6625 Train0.6375 Random State 5
        Test 0.65 Train0.640625 Random State 6
        Test 0.675 Train0.634375 Random State 7
        ...
        Test 0.9125 Train0.834375 Random State 286
```

```
In [8]: x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state
=286)
        finalModel=LogisticRegression()
        finalModel.fit(x_train,y_train)
Out[8]: LogisticRegression()

In [9]: print(finalModel.score(x_train,y_train))
        print(finalModel.score(x_test,y_test))

        0.834375
        0.9125
```

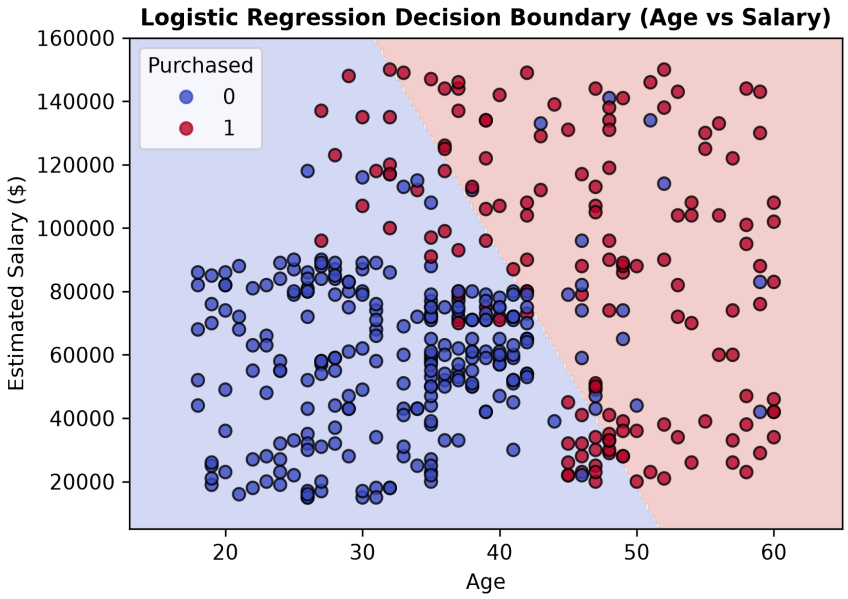
OUTPUT: 3. Classification Metrics & Decision Boundary Plot

```
In [10]: from sklearn.metrics import classification_report
         print(classification_report(label,finalModel.predict(features)))

                precision    recall  f1-score   support

           0       0.85      0.93      0.89       257
           1       0.84      0.71      0.77       143

    accuracy                           0.85       400
   macro avg       0.85      0.82      0.83       400
weighted avg       0.85      0.85      0.85       400
```



Result:

Logistic Regression successfully classified customer purchase decisions with approximately 85% accuracy. Feature scaling significantly improved model convergence and performance. The decision boundary plot revealed clear separation between purchasing and non-purchasing customer segments.